

Fourier trans: $\delta^3(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}}$

$\Rightarrow \tilde{\delta}(k) = 1$

$\Delta \equiv \nabla^2$

$\Delta^n \delta^3(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} \Delta^n e^{i\vec{k}\cdot\vec{x}} = \int \frac{d^3k}{(2\pi)^3} (-k^2)^n e^{i\vec{k}\cdot\vec{x}}$

$\therefore \mathcal{F}\{\Delta^n \delta^3(\vec{x})\}(k) = -(k^2)^n$

More generally:

$\mathcal{F}\{\Delta^n f\}(k) = -(k^2)^n \tilde{f}(k)$

* Charge at the origin:

$J^\mu = \begin{cases} J_0(x) = \rho(x) = e\delta^3(x) \\ J_i(x) = 0 \end{cases}$

$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu$

E.O.M $\square A_\mu = J_\mu$

$\Leftrightarrow A_\mu = \frac{1}{\square} J_\mu$

Since $J_i = 0$ we are left with

$A_0 = \frac{1}{\square} (e\delta^3(x)) = \left(\frac{e}{\Delta}\right) (e\delta^3(x))$

But $\delta^3(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}}$

$\therefore A_0(\vec{x}) = e \int \frac{d^3k}{(2\pi)^3} \frac{1}{\Delta} e^{i\vec{k}\cdot\vec{x}}$
 $= e \int \frac{d^3k}{(2\pi)^3} \frac{1}{k^2} e^{i\vec{k}\cdot\vec{x}}$

Work in spherical coordinates $r^2 = x^2 + y^2 + z^2$

$A_0(x) = \frac{e}{(2\pi)^3} \int_0^\infty dk k^2 \int_0^\pi d\cos\theta \int_0^{2\pi} d\phi \frac{1}{k^2} e^{i\vec{k}\cdot\vec{x}}$

$= \frac{e}{(2\pi)^2} \int_0^\infty dk k^2 \int_{-1}^1 du e^{i\vec{k}\cdot\vec{x}}$

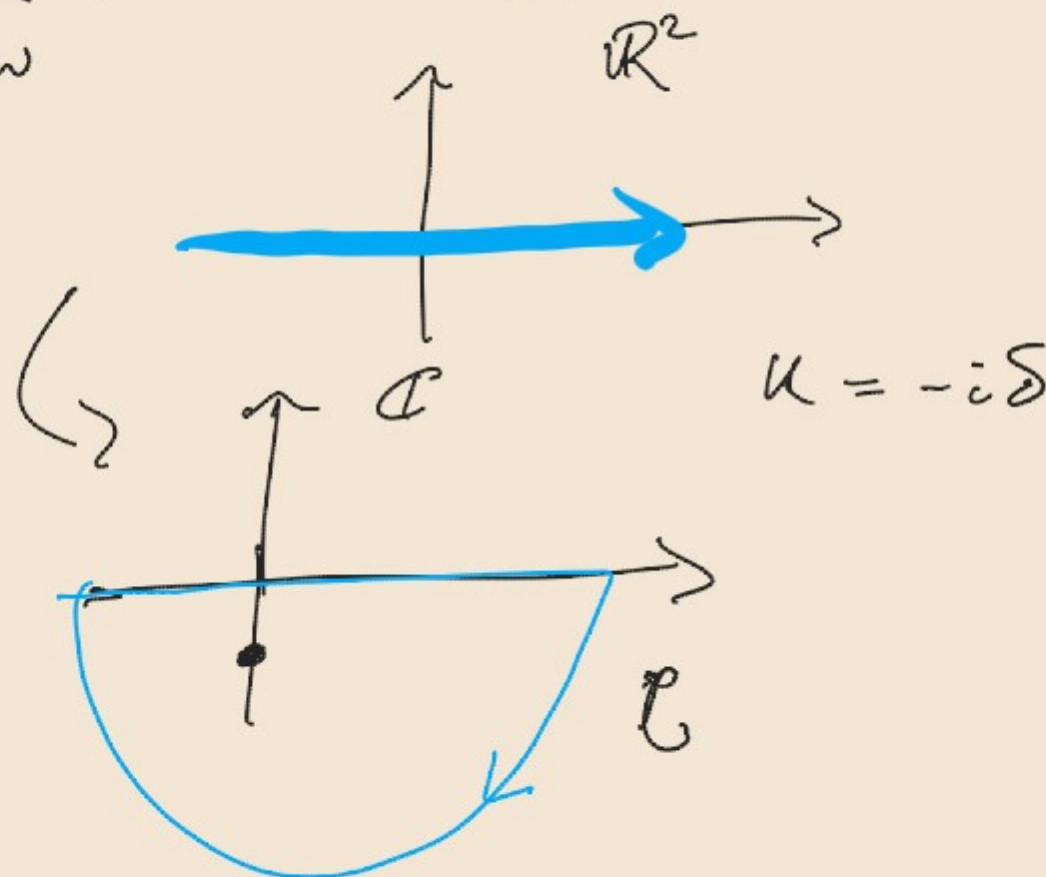
$= \frac{e}{(2\pi)^2} \int_0^\infty dk k^2 \left(\frac{2}{ikr}\right) (e^{ikr} - e^{-ikr})$

$= \frac{e}{(2\pi)^2} \left(\frac{1}{ir^2}\right) \int_{-\infty}^\infty dk \frac{e^{ikr} - e^{-ikr}}{k}$

$= \frac{e}{(2\pi)^2} \left(\frac{1}{ir^2}\right) \ln\left(\frac{e^{ikr} + e^{-ikr}}{e^{ikr} - e^{-ikr}}\right)$

Note: $\lim_{k \rightarrow 0} \left(\frac{1}{k} \right) = \frac{1}{0} = \infty$
 $= 2i\pi$

Now shift the pole $k \rightarrow k + i\delta$
 $\int_{-\infty}^{\infty} \frac{dk}{k} = \lim_{\delta \rightarrow 0} \left(\int_{-\infty}^{\infty} \frac{e^{-k\delta} e^{ikr} - e^{k\delta} e^{-ikr}}{k + i\delta} \right)$
 Now



$\int_{-\infty}^{\infty} dk = \int_C$ in the limit that $R \rightarrow \infty$

$k + i\delta = R e^{i\theta}$

* If $\delta > 0$: For e^{ikr} we close the pole above to get exponential decay

$\int_{-\infty}^{\infty} \frac{e^{ikr}}{k + i\delta} dk = \int_{-\infty}^{\infty} dk \frac{e^{ikr}}{R e^{i\theta}}$

$k + i\delta = R e^{i\theta} \Rightarrow dk = dR e^{i\theta} + R i e^{i\theta} d\theta$
 $\int dR \frac{e^{i\theta} R e^{i\theta} e^{-i\delta r}}{R} + \int d\theta \frac{e^{i\theta} R e^{i\theta} e^{-i\delta r}}{R}$

$$\int_{-\infty}^{\infty} dk = e^{-i\epsilon r} f(k+i\epsilon) = -2\pi i (-e^{\delta r}) = 2\pi i e^{\delta r}$$

$$\therefore \boxed{A_0(x) = \frac{e}{4\pi} \frac{1}{r}}$$

Green's functions:

Lagrangian: $\mathcal{L} = -\frac{1}{2} h \square h + \frac{1}{3} \lambda h^3 + J h$

E.O.M: $\square h - \lambda h^2 - J = 0$

• For $\lambda=0$: $\square h_0 = J$ and $h_0 = \frac{1}{\square} J$

$h = h_0 + h_1$, $h_1 = \mathcal{O}(\lambda^2)$

$\square(h_0 + h_1) - \lambda(h_0 + h_1)^2 + J = 0$

$\square h_1 - \lambda(h_0 + h_1)^2 = 0$

$\square h_1 - \lambda(h_0^2 + 2h_0 h_1 + h_1^2) = 0$

$\therefore \square h_1 = \lambda h_0^2 + \mathcal{O}(\lambda^2)$

and so

$h_1 = \lambda \frac{1}{\square} (h_0 h_0) + \frac{1}{\square} \mathcal{O}(\lambda^2)$

$h_1 = \lambda \frac{1}{\square} \left[\left(\frac{1}{\square} J \right) \left(\frac{1}{\square} J \right) \right]$

Thus, to order λ , the solⁿ is

$h = \frac{1}{\square} J + \lambda \frac{1}{\square} \left[\left(\frac{1}{\square} J \right) \left(\frac{1}{\square} J \right) \right] + \mathcal{O}(\lambda^2)$

$\pi \equiv -\frac{1}{\square}$ is defined as a solution to
 $\square_x \pi(x, y) = -\delta^d(x-y)$, $\square_K \equiv g^{\mu\nu} \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^\nu}$

$F(\square_x \pi) = -\int \delta^d(x-y)$

$= K^2 \pi(K) = -1 \Rightarrow \tilde{\pi}(K) = \frac{1}{K^2}$

$F^{-1}(\tilde{\pi}(K)) = F^{-1}\left(\frac{1}{K^2}\right)$

* Useful expansion:

$$h(x) = \int d^4y \delta^4(x-y) h(y) = - \int d^4y (\Box_y \pi(x,y)) h(y) \\ = - \int d^4y \pi(x,y) \Box_y h(y)$$

Thus, we find that, for $\Box_y h(y) = J(y)$
 $h(x) = - \int d^4y \pi(x,y) J(y)$

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$$\Box_\omega h_2(\omega) = \lambda h_0^2(\omega) = \lambda \int d^4y \pi(\omega,y) J(y) \int d^4z \pi(\omega,z) J(z)$$

$$h(x) = - \int d^4y \pi(x,y) J(y) \\ + \lambda \int d^4\omega \int d^4y \int d^4z \pi(x,\omega) \pi(\omega,y) \pi(\omega,z) J(y) J(z) + \dots$$



Nonlinear gravity as a classical field theory:

$$\mathcal{L} = M_{Pl}^2 \left(-\frac{1}{2} h_{\mu\nu} \Box h^{\mu\nu} + \partial_\alpha h_{\mu\nu} \partial_\beta h^{\mu\alpha} \partial^\nu{}^\beta + \dots \right) \\ - h_{\mu\nu} T^{\mu\nu}$$

$$\mathcal{L} = -\frac{1}{2} h \Box h + (M_{Pl})^2 h^2 \Box h - (M_{Pl})^4 h T$$

• $h \sim \bar{L}$

$E = mc^2$, $p = \frac{h\nu}{\lambda}$, $E = pc$

• $m \sim \frac{1}{L}$

$$h^{\mu\nu} \Box h_{\mu\nu} \sim h^{\mu\nu} (g^{\alpha\beta} \partial_\alpha \partial_\beta) h_{\mu\nu}$$

$$\mathcal{S} = -m \int d^4x \mathcal{L}$$

$$a = b = -1$$

$$E \sim M: \quad \Box h = (M_{Pl})^2 \Box(h^2) = M_{Pl}^4 T$$

Point Source $T = m\delta^{(3)}(\vec{x})$.

• In the massless case we have

$$\square = \partial_t^2 - \Delta_{(3)}$$

$$\begin{aligned} h &= -\frac{1}{\square} M_{Pl}^2 T \\ &= -\frac{M_{Pl}^2}{\square} m \left(\int \frac{d^4 k}{(2\pi)^4} e^{i k x} \right) \\ &= -\frac{M_{Pl}^2 m}{(2\pi)^4} \int dK_0 d\vec{k} = \frac{M_{Pl}^2 m}{(2\pi)^4} \int d^4 k \frac{1}{\Delta} e^{i k x} \\ &= \frac{M_{Pl}^2 m}{(2\pi)^4} \int d^4 k \left(\frac{1}{-k^2} \right) e^{-i k \cdot x} \end{aligned}$$

$$\begin{aligned} &K(x) \cos(\theta) \\ &\frac{M_{Pl}^2 m}{(2\pi)^3} \int d\theta d\phi d\vec{k} \left(\frac{1}{-k^2} \right) e^{i k(x) \cos(\theta)} \end{aligned}$$

$$\begin{aligned} h &= \frac{(M_{Pl}^2 m)}{(2\pi)^3} \int_0^\infty k^2 dk \int_{-1}^1 d\cos(\theta) \int_0^{2\pi} d\phi \frac{1}{k^2} e^{i k r \cos(\theta)} \\ &= \left(\frac{M_{Pl}^2 m}{(2\pi)^2} \right) \int_0^\infty dk e^{i k r} = \frac{e^{i k r}}{i k r} \end{aligned}$$

$$k \rightarrow k + i\delta$$

Pole of $k = -i\delta$, close on Bottom contour

$$\begin{aligned} &k \rightarrow -\infty \quad \text{so } e^{i k r} \rightarrow 0 \\ &\lim_{\delta \rightarrow 0} \int_{-\infty}^{\infty} dk \left(\frac{e^{-i k r}}{i k} \right) = (2\pi i) \frac{1}{i r} e^{-\frac{1}{2}(i\delta)r} \frac{1}{(2\pi)^2} \end{aligned}$$

$$= \left(\frac{M_{Pl}^2 m}{(2\pi)^2} \right) \left(\frac{1}{i r} \right) 2\pi i e^{\delta r} = -\frac{M_{Pl}^2 m}{4\pi r}$$

$$h = \frac{M_{Pl}^2 m}{4\pi r}$$

$$h = h^{(1)} + h^{(2)}$$

$$\square(h^{(1)} + h^{(2)}) = M_{Pl}^2 (h^{(1)} + h^{(2)})^2 + M_{Pl}^2 T$$

- $\square h^{(2)} = M_{pe}^{-1} \square (h^{(1)})^2$

or

$$h^{(2)} \approx M_{pe}^{-1} (h^{(1)})^2 = \frac{M_{pe}^{-3} m^2}{r^2}$$

$$\omega = \frac{2\pi}{T}$$

Newtonian potential: $\phi = M_{pe}^{-1} h^{(1)}$

$$\phi = Gm/4\pi r$$

$$-\frac{\partial \phi}{\partial r} = U, \text{ potential energy}$$



$$v = \omega R$$

$$\int F dx = T$$

$$M \propto R = \frac{1}{2} m v^2$$

Factor of 2?

$$\alpha = \frac{v^2}{R}$$

$$\alpha = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx} = \omega R \frac{d\omega}{dx}$$

~~$$\omega = \left(\frac{2\pi}{T} \right) \frac{dt}{dx} = \omega \frac{1}{T} \frac{1}{\frac{dx}{dt}} = \frac{\omega}{T \omega R}$$~~

$$\alpha = \omega^2 R = \left| \frac{\partial^2 \phi}{\partial r^2} \right|$$

$$\omega^2 = G N M_{sun} / R$$

$$h^{(2)} = M_{pe}^{-3} m^2 / r^2$$

$$F \propto \frac{G m^2}{r^2}$$

